

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 10%

QUESTIONS: 05 Total Marks:100 Annexure A

Comparative Analysis

Code of Conduct:

*I Prudhvi Raj.B(192324274) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:prudhvi raj.b

Annexure A

1. Compare K-Nearest Neighbour and Case-Based Learning using a real-world example such as medical diagnosis. Analyze differences in memory usage, similarity measures, and prediction strategy.
2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.
3. Compare Naïve Bayes and Logistic Regression for a text classification task such as spam detection. Analyze assumptions about feature independence, probability estimation, computational efficiency, and performance on high-dimensional data. Discuss scenarios where one method outperforms the other.
4. Analyze the differences between Support Vector Machines (SVM) and KNN in classification tasks. Use an example dataset to compare decision boundaries, scalability, sensitivity to noise, and computational complexity. Discuss how kernel functions in SVM improve performance for non-linear problems.
5. Compare K-Means Clustering and Hierarchical Clustering using a dataset with unknown structure. Analyze differences in cluster formation, computational cost, scalability, and interpretability. Discuss how each method handles varying cluster shapes and the impact of choosing the number of clusters.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	15	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	20	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. KNN vs Case-Based Learning – Medical Diagnosis

Factor	KNN	Case-Based Learning
Basic idea	Classifies using nearest training samples	Retrieves similar past cases and reuses their solutions
Memory usage	Stores the complete training dataset	Stores previous cases, including solutions/outcomes
Similarity	Usually Euclidean, Manhattan, or other distance measures	Can use weighted/custom similarity measures
Prediction	Majority voting among K neighbours	Uses the closest/relevant case to suggest a solution
Learning	Very little training	Learns by adding new cases
Medical example	Finds K patients with similar symptoms	Finds a previous patient with similar symptoms and diagnosis
Adaptation	New data can be added directly	New cases and their outcomes can be stored for future use

Example:

For a new patient, KNN may find the 5 most similar patients and use majority voting to classify the patient's risk. Case-Based Learning retrieves the most relevant previous medical cases and uses their diagnosis/treatment as a reference.

Conclusion: KNN is simpler and effective for classification, while Case-Based Learning is more suitable when previous cases contain useful solutions that can be reused.

2. Locally Weighted Regression vs RBF Network

Factor	Locally Weighted Regression	RBF Network
Local	Builds a local model for	Uses fixed/learned radial basis

ity	each query	centers
Training	Little or no global training	Requires training of centers and output weights
Prediction	Computationally expensive for each query	Usually faster after training
Function	Local regression	Non-linear function approximation
Smoothness	Depends strongly on bandwidth	Depends on Gaussian width
Memory	Needs training data during prediction	Mainly stores learned parameters
Scalability	Poorer for very large datasets	Better after training
Non-linearity	Captures local non-linearity	Naturally models complex non-linear relationships

Example: Predicting machine temperature from sensor readings.

- LWR gives greater importance to observations close to the current sensor conditions.
- RBF learns representative centers and uses Gaussian activation functions to approximate the entire function.

Conclusion: LWR is useful when local behaviour is important and the dataset is relatively small. RBF networks are preferable when repeated predictions and complex non-linear relationships are required.

3. Naïve Bayes vs Logistic Regression – Spam Detection

Factor	Naïve Bayes	Logistic Regression
Type	Probabilistic classifier	Discriminative classifier
Main assumption	Features are conditionally independent	No feature-independence assumption
Probability	Directly estimates class probabilities	Estimates probability using logistic function
Training speed	Very fast	Generally fast

High-dimensional data	Excellent	Excellent
Text classification	Very effective	Very effective
Data requirement	Works well with relatively small datasets	Often benefits from more training data
Feature relationship	Assumes independence	Can model feature weights jointly
Interpretability	Easy to understand probabilities	Feature coefficients are interpretable

For spam detection, words such as:

"free", "offer", "winner", "money"

can be used as features.

Naïve Bayes

Assumes that the occurrence of each word is conditionally independent given the class.

Logistic Regression

Learns a weight for each word and combines the weights to determine whether the email is spam.

Naïve Bayes is preferable when:

- Training must be extremely fast.
- The dataset is small.
- There are many text features.

Logistic Regression is preferable when:

- Feature relationships matter.
- Sufficient training data is available.
- A strong linear classifier is required.

Conclusion: Both work well for high-dimensional text data. Naïve Bayes is generally simpler and faster, while Logistic Regression can achieve better performance when its assumptions better match the data.

4. SVM vs KNN

Factor	SVM	KNN
Learning type	Supervised	Supervised

Training	Computationally more expensive	Very little training
Prediction	Usually fast after training	Can be expensive
Decision boundary	Explicitly learned	Determined by neighbouring points
Noise sensitivity	Can be controlled using regularization	Sensitive to noisy neighbours
Scalability	Good with appropriate implementations	Poorer for very large datasets
Feature scaling	Important	Very important
Non-linear problems	Kernel functions can help	Naturally handles non-linear boundaries
Memory	Stores model parameters/support vectors	Stores training data
Main parameter	C, kernel, gamma	K, distance metric

Example

For classifying two types of flowers:

KNN:

```

New Flower
  ↓
Find nearest flowers
  ↓
Majority vote
  ↓
Class

```

SVM:

```

Training Data
  ↓
Find optimal separating boundary
  ↓
Maximum margin
  ↓
Classify new flower

```

SVM Kernels

SVM can use kernels such as:

- Linear
- Polynomial
- RBF

The **RBF kernel** allows SVM to create non-linear decision boundaries without explicitly transforming the original features into a higher-dimensional space.

Conclusion: KNN is simple and useful for small datasets, while SVM is generally more suitable when a well-defined decision boundary and strong generalization are required.

5. K-Means vs Hierarchical Clustering

Factor	K-Means	Hierarchical Clustering
Type	Partitioning	Hierarchical
Number of clusters	Usually must specify K beforehand	Can choose clusters after building hierarchy
Output	Separate clusters	Dendrogram/tree
Computational cost	Generally lower	Generally higher
Scalability	Good for large datasets	Less suitable for very large datasets
Cluster shape	Works best with roughly spherical clusters	Can represent more varied structures depending on linkage
Initialization	Sensitive to initial centroids	No centroid initialization
Interpretability	Simple	Dendrogram provides hierarchy
Outlier sensitivity	Sensitive	Depends on linkage method
Reassignment	Points can change clusters during iterations	Merges/splits are generally irreversible

K-Means

Data
↓
Choose K
↓
Initialize centroids

↓
Assign points
↓
Update centroids
↓
Repeat
↓
Final clusters

Hierarchical Clustering

Individual points
↓
Find closest clusters
↓
Merge clusters
↓
Repeat
↓
Dendrogram
↓
Choose desired clusters

Cluster shape

K-Means works best when clusters are relatively compact and separated.

Hierarchical clustering can discover more varied structures depending on the distance metric and linkage method, although it is not universally better for arbitrary shapes.

Choosing the number of clusters

For K-Means, the number of clusters **K** must normally be selected before training.

For hierarchical clustering, you can inspect the **dendrogram** and select the desired number of clusters afterward.

Conclusion: K-Means is generally faster and more scalable, while hierarchical clustering provides richer structural information and is useful when the underlying number of clusters is initially unknown.